$^{34}S(n,n),(n,\gamma)$:resonances 1984Ca14

1984Ca14: 30-1500-keV neutrons were produced from the Oak Ridge Electron Linear Accelerator (ORELA). The target was 94.33%-enriched ^{34}S with a 5.54% impurity of ^{32}S. Neutrons were detected using a 2-cm-thick and 7.5-cm in diameter NE110 proton-recoil detector. Prompt γ rays from neutron capture were detected using two nonhydrogenous fluorocarbon scintillators (NE-226). Measured $\sigma(E_n)$. Deduced 81 resonances, J, π, L, widths and spectroscopic factors by comparing the R-matrix predictions of the cross section with the data.

2018MuZY: Atlas of Neutron Resonances. Evaluation of neutron resonance energies, spins, width parameters, and resonance strengths for nuclei of Z=1-60.

^{35}S Levels

J^π and L are from 2018MuZY. Resonance parameters E_n, $g\Gamma_n$, Γ_γ, and $g\Gamma_n\Gamma_\gamma/\Gamma$ are from 1984Ca14 due to significant discrepancies, including orders of magnitude inconsistencies, between the 1984Ca14 original data and 2018MuZY evaluation. Here, $g\Gamma_n=(2J+1)\Gamma_n/2$.

E(level)†	J^π	$g\Gamma_n\Gamma_\gamma/\Gamma$	L	E_n(lab) (keV)	Comments
6985.54? 4	1/2		0	−0.305	E(level): fictitious level from analysis of resonance spectrum below the S(n) threshold. Γ_γ=0.38 eV.
7018.89 4	1/2	0.025 eV 1	[2]	34.03 1	
7072.66 5		0.14 eV 4		89.40 3	
7074.33 6		0.084 eV 3		91.12 4	
7097.72 29		0.009 eV 3	[2]	115.2 3	
7099.8 2		0.033 eV 5	[2]	117.4 2	$g\Gamma_n$<1 eV.
7100.73 4	3/2		[1]	118.30 1	$g\Gamma_n$=372 eV 3; Γ_γ=0.70 eV 4.
7142.4 1		0.42 eV 1	[2]	162.1 1	$g\Gamma_n$<1 eV.
7210.42 5			[1]	231.25 2	$g\Gamma_n$=62 eV 4; Γ_γ=0.30 eV 2.
7217.9 4		0.33 eV 2	[2]	239.0 4	$g\Gamma_n$<4 eV.
7233.9 5		0.19 eV 1	[2]	255.5 5	$g\Gamma_n$<4 eV.
7239.8 6		0.06 eV 2	[2]	261.5 6	$g\Gamma_n$<2 eV.
7253.2 6		0.35 eV 2	[2]	275.3 6	$g\Gamma_n$<1 eV.
7275.93 5	1/2		0	298.70 3	$g\Gamma_n$=9600 eV 150; Γ_γ=2.82 eV 67.
7279 3		0.10 eV 3		302	
7289.8 6		0.31 eV 3	[2]	313.0 8	$g\Gamma_n$<6 eV.
7294.19 7	3/2		1	317.51 5	$g\Gamma_n$=2500 eV 25; Γ_γ=0.23 eV 8.
7331.00 5	1/2		0	355.41 2	$g\Gamma_n$=4810 eV 80; Γ_γ=0.30 eV 30.
7338.1 2			[1]	362.8 2	$g\Gamma_n$=230 eV 10; Γ_γ=1.06 eV 7.
7343.9 2		0.72 eV 3	[2]	368.7 2	$g\Gamma_n$<2 eV.
7347.2 5		0.44 eV 6	[2]	372.1 5	$g\Gamma_n$<2 eV.
7347.9 5		0.30 eV 5	[2]	372.8 5	$g\Gamma_n$<2 eV.
7353.9 5		0.44 eV 4	[2]	379.0 5	$g\Gamma_n$<10 eV.
7357.6 5		0.16 eV 4	[2]	382.8 5	$g\Gamma_n$<6 eV.
7370.52 6	1/2		1	396.14 4	$g\Gamma_n$=6430 eV 150; Γ_γ=3.08 eV 20.
7372 4		0.12 eV		397.6	
7395.9 6		0.44 eV 33	[2]	422.3 6	
7404.6 6		0.21 eV 5	[2]	431.2 6	$g\Gamma_n$<1 eV.
7408.6 1	3/2		1	435.4 1	$g\Gamma_n$=1570 eV 40; Γ_γ=0.91 eV 8.
7411.55 7			[2]	438.35 5	$g\Gamma_n$=49 eV 5.
7416.50 4			[2]	443.45 1	$g\Gamma_n$=80 eV 8; Γ_γ=0.51 eV 5.
7429.0 6		0.12 eV 3	[2]	456.4 6	$g\Gamma_n$<6 eV; Γ_γ=0.12 eV 3.
7433.56 5			[2]	461.01 2	$g\Gamma_n$=50 eV 5; Γ_γ=0.035 eV 20.
7437 4			[1]	463.9 4	$g\Gamma_n$=296 eV 30; Γ_γ=0.28 eV 3.
7442.14 5	1/2		0	469.85 2	$g\Gamma_n$=1050 eV 35; Γ_γ=0.37 eV 8.
7462.2 6		0.33 eV 4	[2]	490.5 6	$g\Gamma_n$=50 eV 10.
7481.33 6	1/2		1	510.02 4	$g\Gamma_n$=375 eV 25; Γ_γ=2.1 eV 1.
7494.5 6		0.17 eV 4	[2]	523.8 6	$g\Gamma_n$<4 eV.

Continued on next page (footnotes at end of table)

^{34}S(n,n),(n,γ):resonances 1984Ca14 (continued)

^{35}S Levels (continued)

E(level)†	J^π	$g\Gamma_n\Gamma_\gamma/\Gamma$	L	E_n(lab) (keV)	Comments
7542.3 *6*		0.21 eV *3*	[2]	573.0 *6*	$g\Gamma_n$<1 eV.
7603.9 *6*		0.90 eV *8*	[2]	636.5 *6*	$g\Gamma_n$<10 eV.
7609.26 *5*	3/2		1	641.93 *3*	$g\Gamma_n$=2520 eV *60*; Γ_γ=0.51 eV *6*.
7648.8 *6*		0.21 eV *3*	[2]	682.7 *6*	$g\Gamma_n$<30 eV.
7655.6 *6*		0.80 eV *5*	[2]	689.7 *6*	$g\Gamma_n$<10 eV.
7663.9 *1*			[2]	698.2 *1*	$g\Gamma_n$=305 eV *50*; Γ_γ=0.80 eV *8*.
7678.2 *7*		0.45 eV *7*	[2]	713.0 *7*	$g\Gamma_n$<4 eV.
7731.0 *6*		0.31 eV *13*	[2]	767.3 *7*	$g\Gamma_n$<4 eV.
7749.6 *4*	(3/2)		[1]	786.5 *4*	$g\Gamma_n$=18 eV *6*; Γ_γ=2.65 eV *18*.
7761.5 *6*	3/2		1	798.65 *6*	$g\Gamma_n$=25350 eV *300*; Γ_γ=1.52 eV *50*.
7776.1 *1*	1/2		1	813.8 *1*	$g\Gamma_n$=860 eV *60*; Γ_γ=0.19 eV *13*.
7797.99 *9*	1/2		0	836.27 *8*	$g\Gamma_n$=3600 eV *160*; Γ_γ=0.39 eV *18*.
7812.24 *7*	3/2		2	850.94 *6*	$g\Gamma_n$=2650 eV *110*; Γ_γ=0.24 eV *19*.
7853.25 *7*	3/2		1	893.17 *6*	$g\Gamma_n$=4950 eV *160*; Γ_γ=0.38 eV *16*.
7861.8 *12*		1.13 eV *19*	[2]	902.0 *12*	$g\Gamma_n$<3 eV.
7880.7 *12*		0.67 eV *17*	[2]	921.5 *12*	$g\Gamma_n$<20 eV.
7889.0 *12*		0.57 eV *25*	[2]	930.0 *12*	$g\Gamma_n$<20 eV.
7894.63 *7*	3/2		1	935.78 *6*	$g\Gamma_n$=4100 eV *160*; Γ_γ=3.48 eV *23*.
7899.73 *7*			[2]	941.0 *3*	$g\Gamma_n$=176 eV *16*; Γ_γ=0.5 eV *3*.
7934.0 *15*		0.5 eV *3*	[2]	976.4 *15*	$g\Gamma_n$<2 eV.
7939.5 *15*		0.89 eV *4*	[2]	982.0 *15*	$g\Gamma_n$<8 eV.
7954.96 *7*	3/2		1	997.9 *1*	$g\Gamma_n$=930 eV *80*; Γ_γ=3.51 eV *19*.
7974.2 *1*	3/2		1	1017.7 *1*	$g\Gamma_n$=4020 eV *180*; Γ_γ=1.22 eV *17*.
8019.5 *2*			[2]	1064.4 *2*	$g\Gamma_n$=1032 eV *90*.
8041.2 *2*			[2]	1086.7 *2*	$g\Gamma_n$=2650 eV *160*.
8076.4 *2*			[2]	1123.0 *2*	$g\Gamma_n$=5480 eV *388*.
8077.9 *3*			[2]	1124.5 *3*	$g\Gamma_n$=4730 eV *320*.
8092.9 *5*	1/2		1	1140.0 *5*	$g\Gamma_n$=34880 eV *830*.
8143.4 *2*			[2]	1192.2 *2*	$g\Gamma_n$=5510 eV *290*.
8187.17 *3*	1/2		1	1237.0 *3*	$g\Gamma_n$=2945 eV *230*.
8211.25 *3*			[1]	1261.8 *3*	$g\Gamma_n$=975 eV *86*.
8224.1 *4*	1/2		1	1275.1 *4*	$g\Gamma_n$=2170 eV *180*.
8228.3 *4*			[2]	1279.4 *4*	$g\Gamma_n$=1140 eV *100*.
8243.98 *3*			[2]	1295.5 *3*	$g\Gamma_n$=1980 eV *150*.
8256.3 *2*	3/2		1	1308.2 *2*	$g\Gamma_n$=12475 eV *770*.
8262.4 *4*	5/2		2	1314.5 *4*	$g\Gamma_n$=6240 eV *500*.
8273.3 *4*	1/2		1	1325.7 *4*	$g\Gamma_n$=1635 eV *145*.
8298.27 *3*			[2]	1351.4 *3*	$g\Gamma_n$=14470 eV *650*.
8333.8 *4*	3/2		1	1388.0 *4*	$g\Gamma_n$=28845 eV *1057*.
8335.8 *4*			[2]	1390.1 *4*	$g\Gamma_n$=1955 eV *170*.
8391.3 *4*	3/2		1	1447.2 *4*	$g\Gamma_n$=27100 eV *1390*.
8393.6 *3*			[2]	1449.6 *3*	$g\Gamma_n$=3025 eV *260*.
8406.55 *3*			[2]	1462.9 *3*	$g\Gamma_n$=3885 eV *345*.
8418.79 *4*			[2]	1475.5 *4*	$g\Gamma_n$=3795 eV *340*.

† From $E_{c.m.}$+S(n) where S(n)=6985.84 *4* (2021Wa16) and $E_{c.m.}=E_n(\text{lab})\times m(^{34}S)/[m_n+m(^{34}S)]$.